Database Key Concepts

Authored by: Arpit Raj, Lnmiit jaipur

Super Key

Definition: A Super Key is any attribute or set of attributes that uniquely identifies every tuple (row) in a relation.

In terms of Functional Dependencies:

`X` is a Super Key if `X⁺` contains all attributes of the relation. A Super Key may contain unnecessary attributes; it's not minimal.

Candidate Key

A Candidate Key is the smallest (minimal) Super Key. It satisfies two properties:

- **Uniqueness:** Determines all attributes.
- **Irreducibility:** No attribute can be removed.

Attributes Classification

Prime Attributes

Definition: A Prime Attribute is an attribute that belongs to at least one Candidate Key.

Non-prime Attributes

Definition: A Non-prime Attribute is an attribute that is not part of any Candidate Key.

Algorithm to Find Candidate Keys

Given: Relation R, Set of Functional Dependencies F

1. **Step 1:** List all attributes.
2. **Step 2:** Identify attributes that never appear on the RHS of any FD. These must be included in every Candidate Key.
3. **Step 3:** Compute the closure of these attributes.
4. **Step 4:** If the closure contains all attributes, you have a Candidate Key. Otherwise, add more attributes and repeat.
5. **Step 5:** Check minimality. Remove each attribute one at a time.

Example

Relation: R(A, B, C)

FDs: A → B, B → C

Find Candidate Key:

Compute: A^+

Start: {A}

Apply A → B: {A, B}

Apply B → C: {A, B, C}

Candidate Key: A